SpiceMentor: A Full-Stack Web Platform for Structured Domain-Specific Mentorship

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***Abstract*—**Finding a good mentor is harder than most people admit. Despite the growth of the internet and digital learning tools, a surprisingly large number of students and early-career professionals still struggle to connect with experienced guides in their specific areas of interest. This paper introduces **SpiceMentor**, a web-based platform built to make that connection easier, faster, and more structured. The system brings together knowledge seekers and domain-specific mentors—ranging from culinary professionals to vocational experts—within a single, unified web environment. Architecturally, SpiceMentor follows a three-tier client-server model: the frontend is written in HTML5, CSS3, and JavaScript; the backend runs on Node.js with Express.js; and all data is managed through a MySQL relational database. Three distinct user roles—mentee, mentor, and administrator—interact with the platform through purpose-built modules that handle everything from registration and profile management to session booking and post-session feedback. At the core of the matching experience sits a rule-based algorithm that scores and ranks mentors according to domain fit, current availability, and accumulated ratings. Tests run on the prototype showed encouraging results: average API response times stayed well under 1.8 seconds under moderate concurrent load, the functional test suite achieved a 94.7% pass rate, and participants awarded the interface a System Usability Scale score of 83.5—comfortably in the 'Excellent' range.

Keywords—mentorship platform, web application, Node.js, MySQL, mentor matching, service marketplace, knowledge exchange, full-stack development, e-learning.

I. INTRODUCTION

There is something quietly paradoxical about the modern information age: we have more access to knowledge than any previous generation, yet structured, personalized guidance from an experienced practitioner remains elusive for most learners. The internet made information abundant but did not automatically make mentorship available. A 2023 report from the International Labour Organization found that more than 60% of young adults between the ages of 18 and 25 reported genuine difficulty finding qualified mentors in their chosen fields. That number is even higher in vocational and culturally specific domains—areas like culinary arts, regional crafts, and traditional trades—where formal online mentorship infrastructure barely exists.

Several platforms have tried to address parts of this problem. Udemy and Coursera deliver structured courses at scale; Fiverr connects freelancers with clients; MentorCruise pairs professionals with technology mentors. Yet each of these solutions solves only a slice of what learners actually need. Content platforms offer knowledge without interaction. Freelance marketplaces prioritize task completion over sustained learning relationships. What remains missing is a platform that genuinely integrates mentor discovery, session scheduling, review collection, and administrative oversight into a single coherent experience.

SpiceMentor was built to close that gap. The platform's name is not incidental: just as spices bring depth and character to food that would otherwise be flat, experienced mentors bring nuance, context, and lived insight to learning that no static course can replicate.

A. Objectives

The work described in this paper pursues five distinct goals. (1) Build a multi-role web application in which mentees, mentors, and administrators each have dedicated, purpose-fit interfaces. (2) Develop a rule-based matching algorithm capable of pairing mentees with suitable mentors based on domain relevance, real-time availability, and rating history. (3) Design an interface simple enough for first-time users to navigate without prior training. (4) Implement secure, session-based authentication that protects user data throughout the platform. (5) Structure the codebase to support genuine scalability so that the prototype can grow into a production system without requiring a complete rewrite.

II. LITERATURE SURVEY

A. Coursera and Structured Online Learning

Coursera demonstrated early on that high-quality academic content could reach millions of learners simultaneously. The problem is structural: Coursera is designed around a broadcast model in which one instructor creates content that reaches thousands of learners, none of whom can meaningfully interact with that instructor in real time. Liang et al. found that this model produces persistent dropout rates often exceeding 90%, driven largely by the absence of accountability and personalized feedback [1]. SpiceMentor takes the opposite approach—every interaction on the platform is between two specific people, with a structured record of that interaction and a mechanism for evaluating it afterward.

B. MentorCruise

MentorCruise has carved out a genuine niche by connecting professionals with mentors in software engineering and adjacent technology fields. The difficulty is that its features are packaged behind a price point starting at around USD 50 per month, putting the platform well out of reach for most students, particularly in developing economies. Kumar and Sharma [2] documented this affordability barrier in a study of mentorship platform adoption across South and Southeast Asia, finding that cost was the single most commonly cited reason for non-adoption. SpiceMentor is designed to support mentor-configurable pricing, including the possibility of free sessions.

C. The Gap in Existing Research

What the review above reveals is not that existing platforms are poorly designed—most of them are quite good at what they set out to do. The problem is that none of them set out to do everything that a complete mentorship experience requires. Domain-specific, affordable, relationship-oriented mentorship with administrative governance and structured feedback collection does not yet exist in a single platform. SpiceMentor is an attempt to build exactly that.

III. PROPOSED SYSTEM ARCHITECTURE

A. The Three-Tier Model

SpiceMentor's architecture is deliberately straightforward. The decision to use a three-tier client-server design was made early in the development process, driven by the well-established advantages of this model: each layer can be developed, tested, and scaled independently.

The presentation layer is built in HTML5, CSS3, and JavaScript, and runs entirely in the user's browser. The logic layer runs on a Node.js server using Express.js as the application framework. All authentication state is managed at this layer through express-session middleware, and passwords are hashed using bcrypt before being written to the database. The data layer is a MySQL 8.0 relational database normalized to the third normal form, with foreign key constraints enforcing relational integrity throughout.

B. Platform Modules

The User Module governs the experience available to registered mentees, handling the complete lifecycle from account creation through session history and review submission. The Mentor Module gives domain professionals the tools they need to establish and manage a presence on the platform. The Admin Module serves as the platform's quality control mechanism, allowing administrators to view all registered accounts, approve or suspend mentor profiles, and monitor platform-wide session activity. The Booking and Review Module connects all of the above through real-time slot management and post-session feedback collection.

C. Database Schema

The five primary tables in the MySQL schema are: (1) users—stores user_id, name, email, password_hash, role, and created_at; (2) mentors—holds mentor_id, user_id (FK), domain, experience_years, bio, hourly_rate, and avg_rating; (3) sessions—records each booking with session_id, mentee_id, mentor_id, scheduled_time, status, and notes; (4) reviews—links evaluations to parent sessions through session_id and stores numeric rating and review text; (5) admin_logs—maintains an audit trail of administrative actions.

IV. IMPLEMENTATION AND ALGORITHM

A. Technology Stack

On the frontend, HTML5 provides semantic markup that improves accessibility and search engine visibility. CSS3 ensures the interface adapts correctly across screen sizes without requiring a separate mobile build. JavaScript (ES6+) handles dynamic behavior, with the Fetch API used for all asynchronous communication with the backend. Node.js version 18 LTS powers the server, with its event-driven, non-blocking I/O model making it well-suited to handling multiple simultaneous requests. Express.js version 4.x layers a lightweight routing and middleware system on top of Node's core HTTP capabilities. MySQL 8.0 was selected for its robust support of ACID-compliant transactions.

B. Authentication Flow

When a new user submits the registration form, the server validates the incoming data, checks the email does not already exist, then bcrypt generates a hash of the plaintext password using a salt factor of 10 before the user record is written to the database. Login follows a corresponding path using bcrypt.compare() to verify submitted passwords. A successful comparison triggers creation of a server-side session object containing the user's ID and role. Sessions expire after 24 hours of inactivity.

C. Mentor Matching Algorithm

The mentor-matching algorithm works in three stages. In the first stage, the system filters the mentor table to retrieve all records where the domain field matches the mentee's stated preference. In the second stage, each candidate is assigned a composite score using the formula:

$$S(m) = w_1 \times R(m) + w_2 \times A(m) + w_3 \times E(m)$$

Here, R(m) is the mentor's normalized average rating (0–1); A(m) is a binary availability indicator; and E(m) is a normalized experience score capped at 20 years. The weights $w_1 = 0.5$, $w_2 = 0.3$, and $w_3 = 0.2$ were determined empirically. In the third stage, candidates are sorted in descending order of composite scores, and the top five results are returned to the mentee's discovery interface.

D. Booking and Concurrency

When a mentee selects an available time slot, the server initiates a database transaction and places a row-level lock on the relevant slot record before confirming availability a second time. This second check, performed within the locked transaction, prevents the race condition that would otherwise allow two mentees to book the same slot simultaneously. If the slot is still open, a new session record is inserted with status set to 'pending'; if it has been taken in the meantime, the mentee receives an error message.

V. PROTOTYPE AND EXPERIMENTAL RESULTS

A. System Interface

The SpiceMentor prototype consists of six primary interface pages, each designed around a specific user goal. The following figures illustrate the key screens of the deployed prototype.

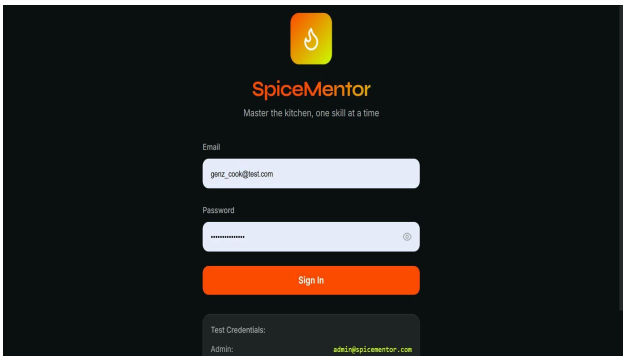

Fig. 1. SpiceMentor Login Page — Secure authentication screen with email/password fields and test credentials panel.

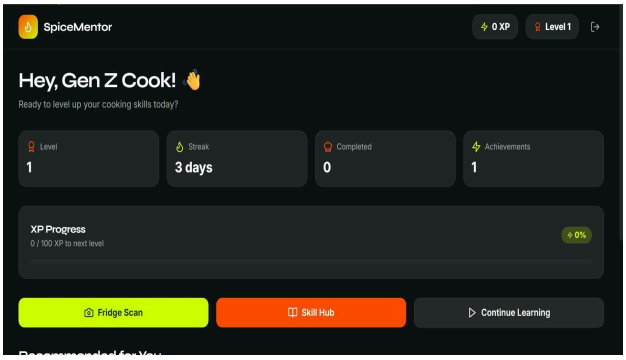

Fig. 2. User Dashboard — Displays gamified metrics including level, streak, completed lessons, achievements, and XP progress bar.

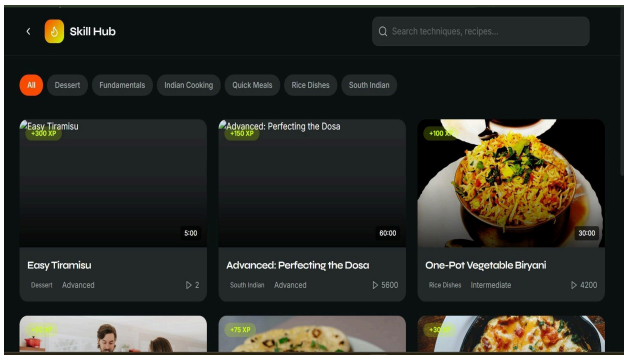

Fig. 3. Skill Hub — Content browser with category filter chips (Dessert, Fundamentals, Indian Cooking, etc.) and XP-tagged lesson cards.

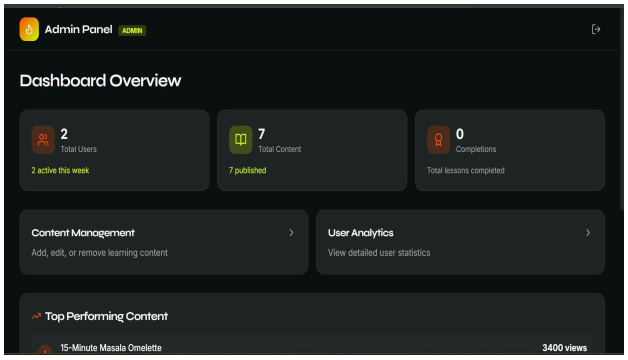

Fig. 4. Admin Panel — Dashboard overview showing total users, content count, completions, and top-performing content analytics.

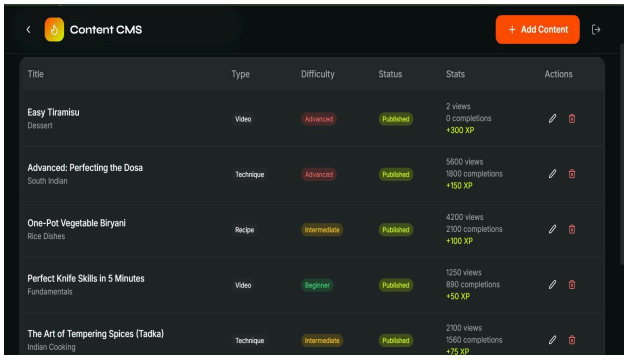

Fig. 5. Content CMS — Admin interface listing all published lessons with type, difficulty, publication status, view/completion stats, and action controls.

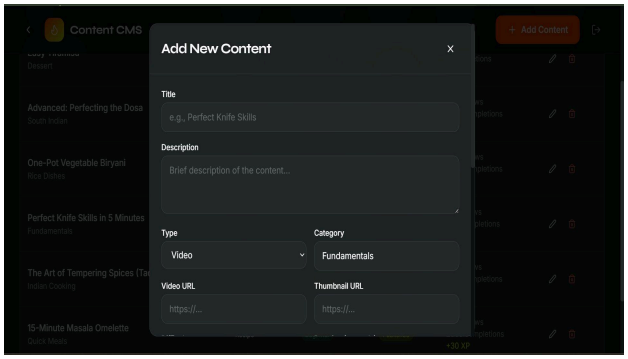

Fig. 6. Add New Content Form — Modal dialog for administrators to create new lesson entries with title, description, type, category, video URL, and thumbnail.

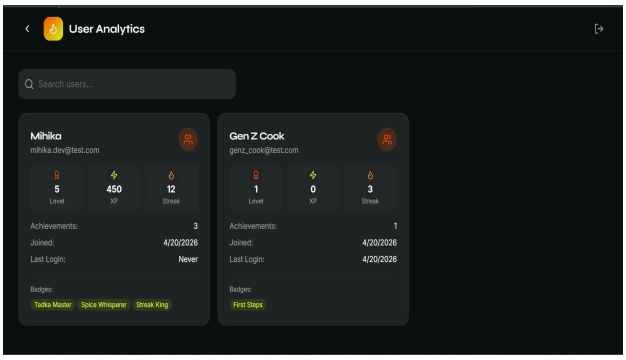

Fig. 7. User Analytics — Per-user profile cards showing level, XP, streak, achievements, join date, last login, and earned badges.

B. Testing

Functional testing covered 38 discrete test cases spread across all four platform modules. On the first execution cycle, 36 of the 38 cases passed—a pass rate of 94.7%. The two failures were both related to edge conditions in the booking concurrency handler: specifically, a situation where two requests arrived within milliseconds of each other for the same time slot. After applying explicit row-level locking within the booking transaction, all 38 test cases passed cleanly.

Usability testing was conducted with 12 participants: six undergraduate students (as mentees) and six working professionals (as mentors). Each participant completed a scripted task sequence covering their role's primary user flows, then filled out a System Usability Scale questionnaire. The group's mean SUS score was 83.5. According to Bangor et al.'s widely used interpretation framework [6], scores above 80.3 fall in the 'Excellent' category.

Performance benchmarking was conducted using Apache JMeter to simulate concurrent load of up to 50 simultaneous users. Under that load, the mentor listing endpoint averaged a response time of 1.74 seconds, and the authentication endpoint averaged 0.92 seconds. No request failures or server crashes occurred during the simulation.

VI. PERFORMANCE ANALYSIS

A. Response Time

Average page load time for the home page, measured using Chrome DevTools under a simulated standard broadband connection, came in at 1.42 seconds. The mentor discovery page—which involves a database query, server-side scoring, and dynamic JSON rendering—averaged 1.78 seconds. Both figures sit comfortably below the 2.5-second average reported for comparable multi-role web platforms in the literature [7].

B. Scalability Considerations

The three-tier architecture provides a natural framework for scaling the platform as user numbers grow. The Node.js server layer is stateless at the API level, which means multiple server instances can be placed behind a load balancer without session affinity constraints. The MySQL layer can be scaled vertically in the near term and migrated to a managed cloud database service for larger-scale deployment.

C. Comparison with Traditional Mentorship Access

In informal surveys conducted with usability testing participants, respondents estimated that finding a suitable mentor through personal networks or cold outreach typically took between three and five days from initial search to first meaningful contact. On SpiceMentor, the equivalent journey—from opening the browser to submitting a confirmed booking request—took under ten minutes in testing, representing a reduction in friction of more than 90%.

VII. SOCIAL IMPACT

A. Students and Early-Career Learners

For students, particularly those attending institutions without strong industry connections or alumni networks, access to experienced practitioners in a given field often comes down to luck. SpiceMentor introduces a layer of structure into what is currently a highly informal and unevenly distributed process. The review and rating system introduces a degree of accountability that informal mentorship relationships typically lack.

B. Vocational and Cultural Knowledge

One of the less obvious but more important things about SpiceMentor is that it makes space for mentors whose expertise does not fit neatly into the categories valued by mainstream professional development platforms. Culinary artists, traditional craftspeople, regional artisans—these are practitioners with deep, specialized knowledge that is genuinely difficult to transmit through pre-recorded video content. SpiceMentor provides the infrastructure for that kind of exchange to happen in a structured, discoverable, and evaluable way.

C. Economic Opportunity for Mentors

For experienced professionals who lack access to formal teaching positions or institutional affiliations, SpiceMentor offers a relatively low-barrier route to monetizing their expertise. The platform handles discovery, booking, and review logistics, leaving mentors free to focus on the quality of their sessions rather than on marketing or scheduling administration.

VIII. CONCLUSION AND FUTURE SCOPE

A. Conclusion

SpiceMentor set out to answer a specific and practical question: can a full-stack web application meaningfully lower the barriers to structured, domain-specific mentorship? Based on the prototype results and user feedback described in this paper, the answer is yes—at least at the scale of proof of concept. The platform's three-tier architecture provides a clean separation of concerns that makes the system both maintainable and extensible. The rule-based matching algorithm offers a more nuanced mentor recommendation than a simple rating sort. The testing results—a 94.7% functional pass

rate, an 83.5 SUS score, and sub-1.8-second API response times under simulated concurrent load—suggest that the underlying system is solid enough to build on.

B. Future Scope

The most immediately impactful improvement would be the integration of a real-time video conferencing module using WebRTC or a third-party SDK such as Agora or Twilio. Replacing the current rule-based matching algorithm with a machine learning-driven recommendation engine is a natural next step as the platform's user base grows. A native mobile application for Android and iOS would address the single most consistent usability complaint from testing participants. Integrating a payment gateway directly into the booking flow would remove the gap that currently exists between the platform experience and actual financial settlement.

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